Chapter 7. Deadlocks. Exercises

Original problems. State assumptions, units, and intermediate reasoning. Answers are in the separate solutions PDF.

OS07-E01. Four necessary conditions

Two processes each hold one exclusive lock and wait for the other lock.

1. Identify the four necessary conditions concretely.
2. Does making one lock preemptible necessarily solve the whole system's deadlock problem?
3. Is a cycle sufficient if another instance can be released?

Check your approach. Necessary conditions organize the analysis; proof of deadlock must exclude the relevant progress transitions.

OS07-E02. Safe and unsafe states

Available=(2,1,0). Allocation is P1=(1,0,0), P2=(0,1,0), P3=(0,0,1). Need is P1=(1,1,0), P2=(2,1,0), P3=(3,0,0).

1. Find a safe completion order.
2. What is Work after P1 completes and returns its allocation?
3. Change P1's Need to (2,1,1). Validate Max against total resources and determine whether the changed state remains safe.

Check your approach. Check 0<=Allocation<=Max<=Total, then show every Work and Finish update.

OS07-E03. Victim selection

A resource is reclaimed while its owner has modified only half the nodes needed for a shared-tree update.

1. Why is merely releasing its lock unsafe?
2. Which recovery unit can restore a consistent state?
3. How can victim selection avoid starvation?

Check your approach. Recovery integrity and victim fairness require separate arguments.

Chapter 7. Deadlocks. Exercises (continued)

OS07-E04 Resource-graph cycle

R1 has two instances. P1 holds one and waits for unique R2. P2 holds R2 and waits for R1. P3 holds the other R1 and needs nothing else.

1. List request and assignment edges.
2. Is there a cycle and do the four conditions appear?
3. Is the state deadlocked? Give a completion order.

OS07-E05 Banker request

T0 Total=(10,5,7), Available=(3,3,2). P0…P4 Allocation=(0,1,0),(2,0,0),(3,0,2),(2,1,1),(0,0,2); Need=(7,4,3),(1,2,2),(6,0,0),(0,1,1),(4,3,1). P1 requests(1,0,2).

1. Check claim and availability componentwise.
2. Give trial Available, Allocation1, Need1.
3. Show every safety Work row and a safe sequence.
4. What must happen if a trial is unsafe?

OS07-E06 Multi-instance detection and recovery

Total=(7,2,6), Available=(0,0,0). P0…P4 Allocation=(0,1,0),(2,0,0),(3,0,3),(2,1,1),(0,0,2); Request=(0,0,0),(2,0,2),(0,0,1),(1,0,0),(0,0,2).

1. Show Work, feasible process, and Finish at every reduction.
2. Identify the final deadlocked set.
3. Abort P1. Show returned resources and remaining reduction.
4. Why does recovery require rollback as well as reclaiming locks?

References and attribution

Abraham Silberschatz, Peter B. Galvin, Greg Gagne. Operating System Concepts, 9th Edition.

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https://github.com/xinhi6716/computer-science-self-study